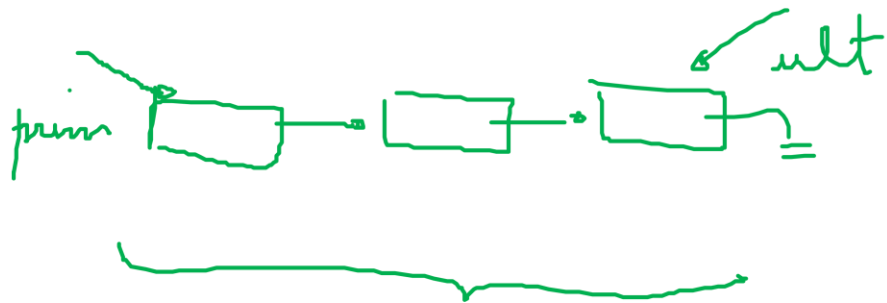


pilha OK
 → operações no início



fila OK

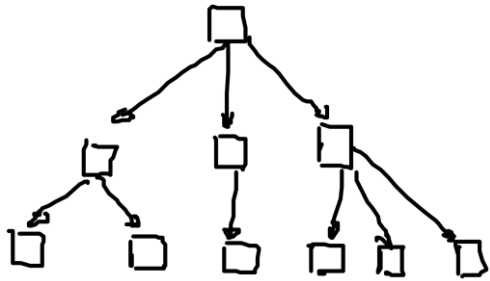
operações de início ^ ^

operações de fim ^ ^

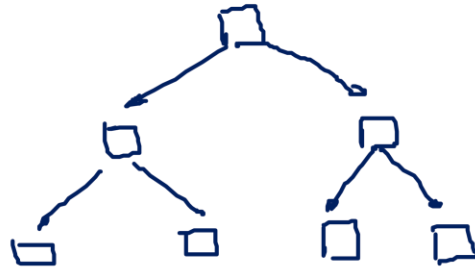
inserções no final

remoções no início

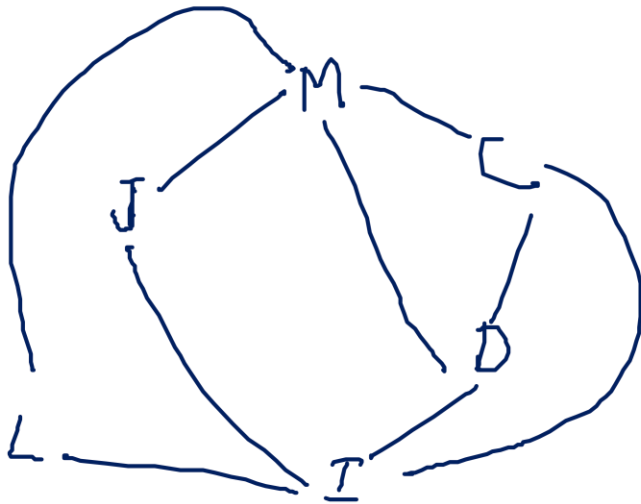
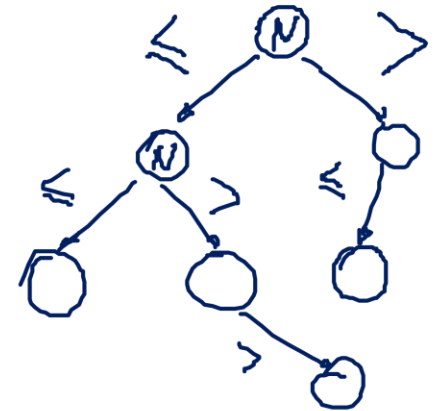
árvores



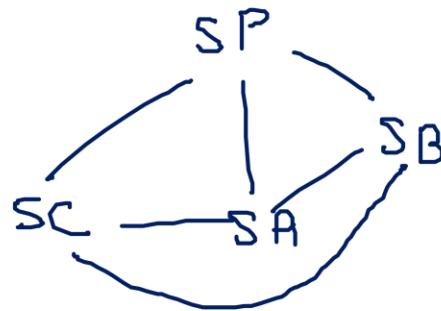
árvores binárias

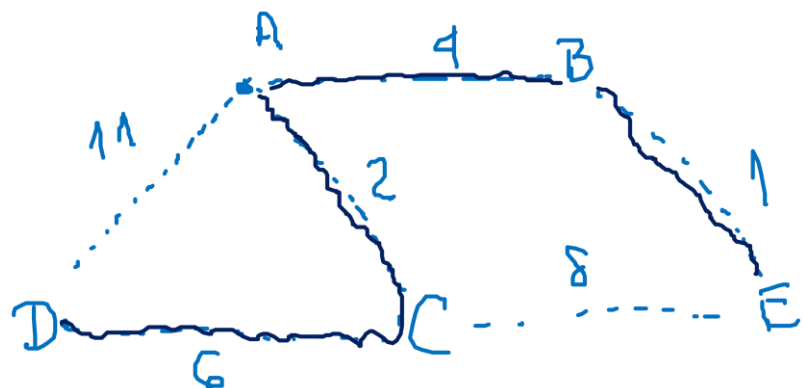


árvores binárias de busca



grafo



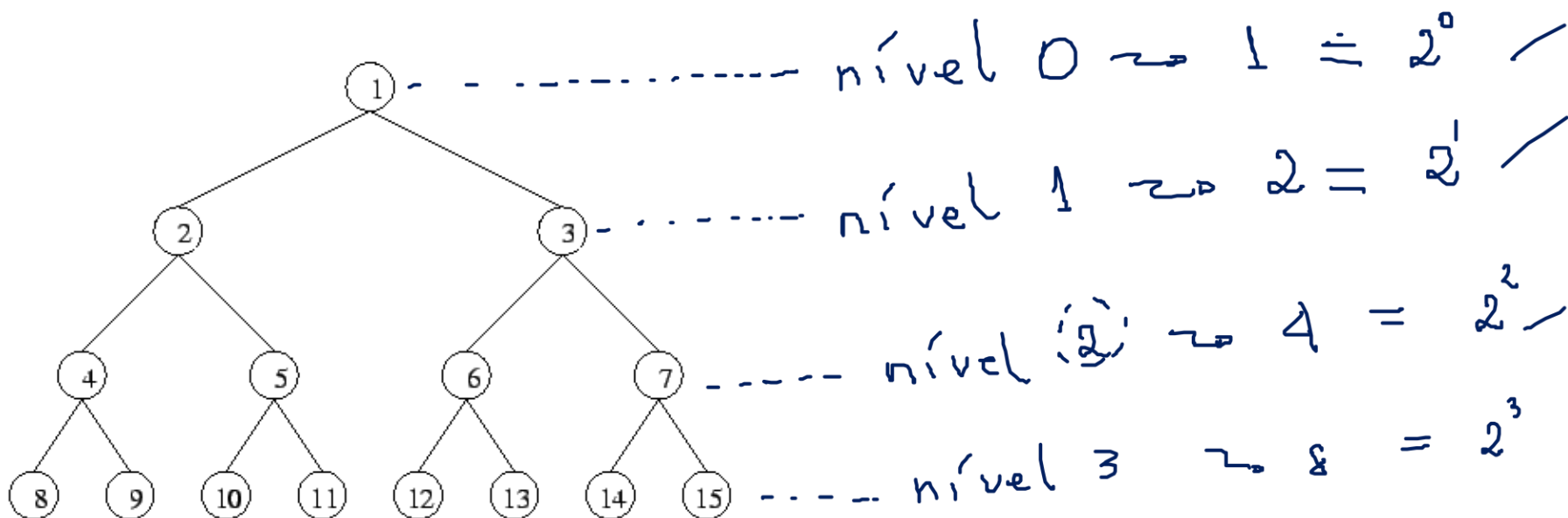


$$(a+b)^0 = 1$$

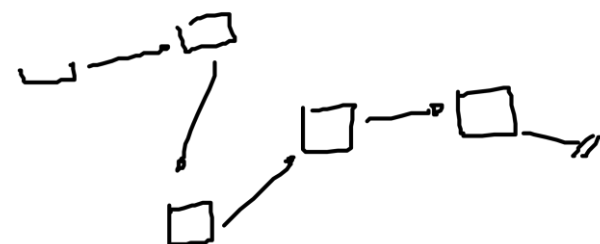
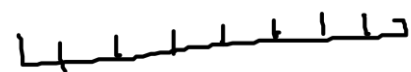
$$(a+b)^1 = 1a + 1b \leadsto s(\text{coef}) = 2$$

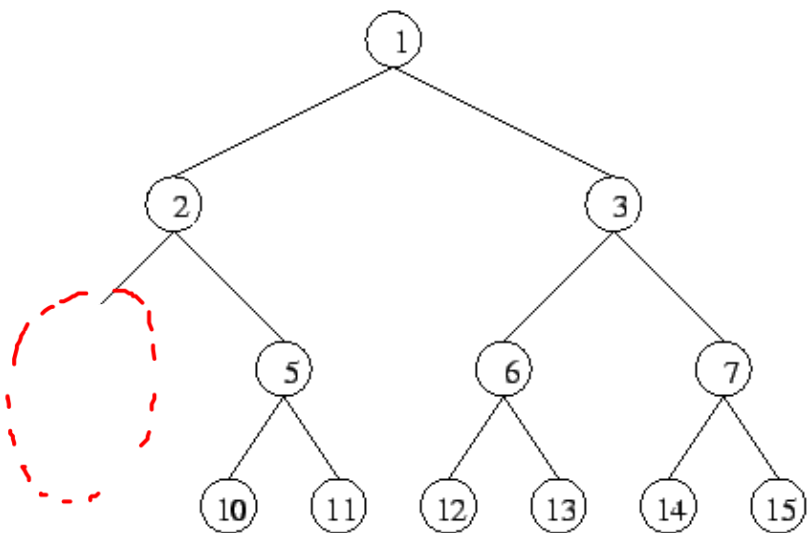
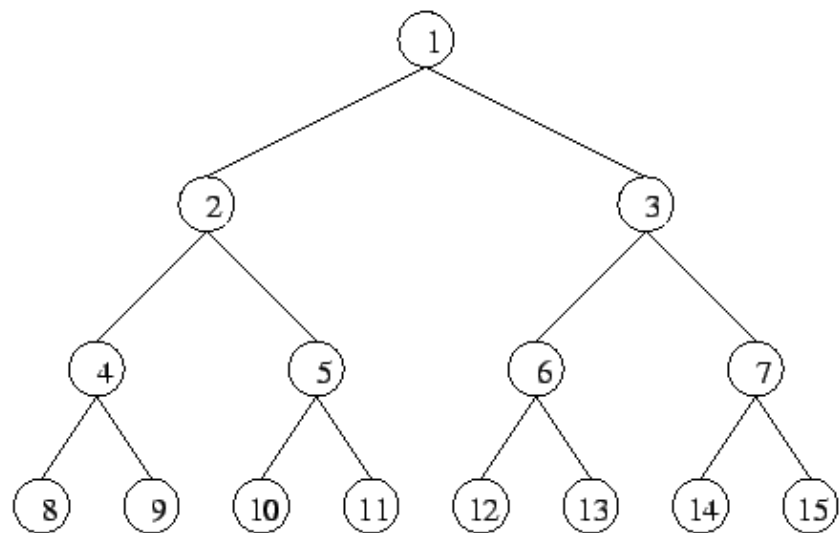
$$(a+b)^2 = 1a^2 + 2ab + 1b^2 \leadsto s(\text{coef}) = 4$$

$$(a+b)^3 = 1a^3 + 3a^2b + 3ab^2 + 1b^3 \leadsto s(\text{coef}) = 8$$



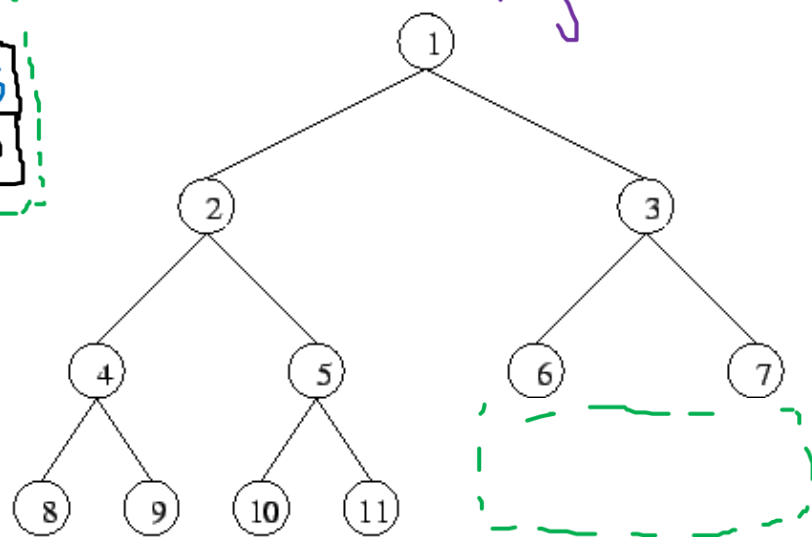
0	1				
1	1	1			
2	1	2	1		
3	1	3	3	1	
4	1	4	6	4	1

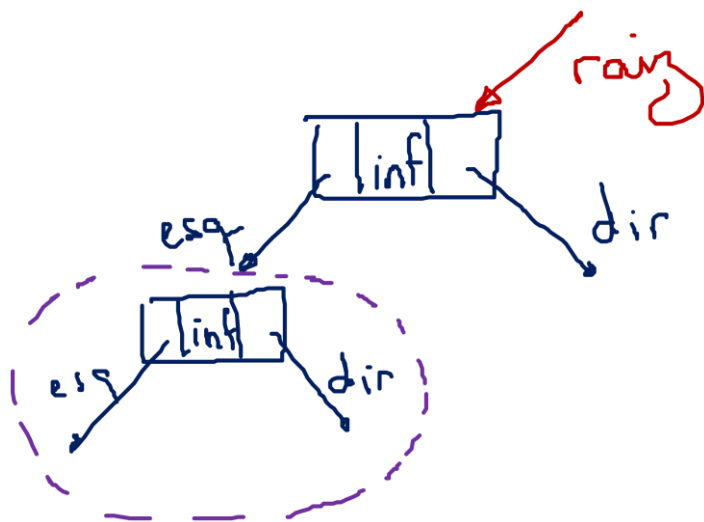




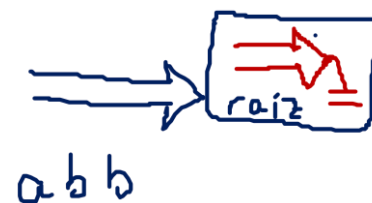
X	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

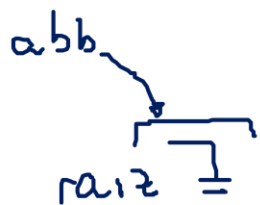
~> heap
~> file priority



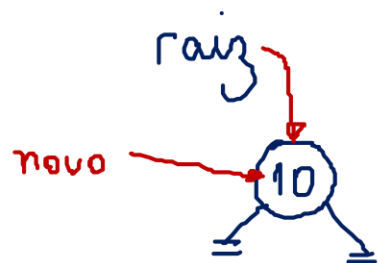


```
void inicia_arvore (t_abb *abb) {  
    abb->raiz = NULL;  
}
```





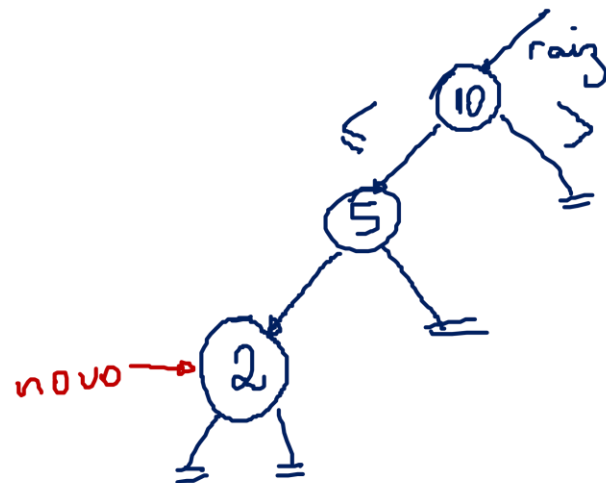
chegou 10



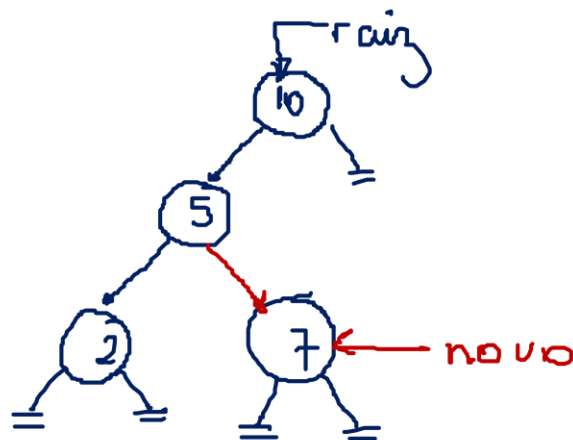
chegou 5



chegou 2



chegou o 7



Nomes

Ana < Bia < Caio

Struct

→ CPF
Name
Endereço

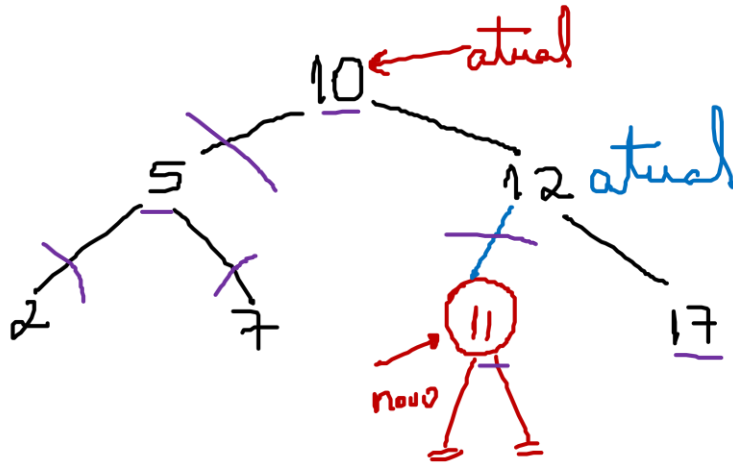
123456 < 567890

```

void insere_rec (t_no *novo, t_no *atual) {
    if (novo->info > atual->info) {
        if (atual->direita == NULL)
            atual->direita = novo;
        else
            insere_rec (novo, atual->direita);
    }
    else {
        if (atual->esquerda == NULL)
            atual->esquerda = novo;
        else
            insere_rec (novo, atual->esquerda);
    }
}

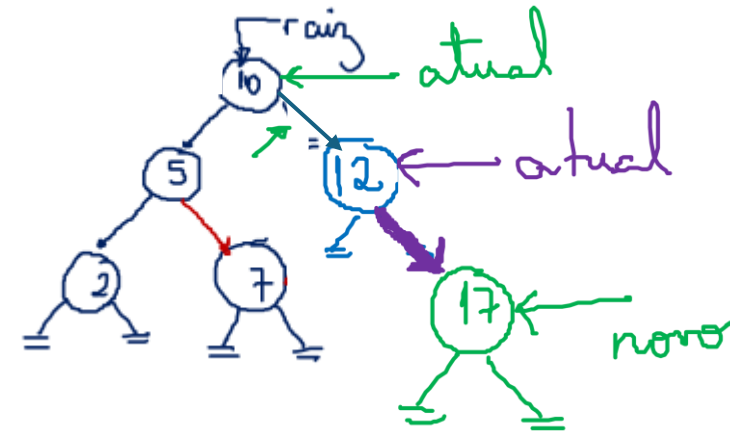
```

cheguei 11



2 5 7 10 11 12 17

cheguei 12



```

void print_em_ordem (t_abb *abb) {
    if (esta_vazia(abb))
        printf ("arvore vazia\n");
    else
        print_em_ordem_rec (abb->raiz);
}

void print_em_ordem_rec (t_no * atual) {
    if (atual->esquerda != NULL)
        print_em_ordem_rec (atual->esquerda);
    printf ("%d ", atual->info);
    if (atual->direita != NULL)
        print_em_ordem_rec (atual->direita);
}

```